# Nicholas Morse

*Curriculum Vitae*

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*Stockholm, Sweden*
*Citizenship: USA*


Researcher with expertise in large-eddy simulation (LES) and direct numerical simulation (DNS) of turbulent boundary layers, jets, and multiphase flows.

## Education

2023 **PhD**, *Aerospace Engineering & Mechanics*, University of Minnesota
*Thesis*: "High-fidelity unstructured overset simulation of complex turbulent flows"
*Adviser*: Professor Krishnan Mahesh

2020 **MS**, *Aerospace Engineering & Mechanics*, University of Minnesota

2018 **BAEM**, *Aerospace Engineering & Mechanics*, University of Minnesota
*Minors*: Math, Astrophysics

## Skills

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|---|---|---|---|
| *Coding* | Fortran, Python, Matlab, Bash, Git | *CFD* | Neko, OpenFOAM, ANSYS, Basilisk |
| *HPC* | MPI, Make, CMake, Hypre | *Meshing* | Pointwise, GridPro, Salome, Gmsh |
| *Visualization* | ParaView, Tecplot, VisIt, PyVista | *Engineering* | SolidWorks, Simulink |

## Experience

May 2025 – Present **Postdoctoral Researcher**, *KTH Royal Institute of Technology*, Stockholm, Sweden
- Performed >1 billion point DNS with the GPU-accelerated spectral element code *Neko*.
- Analyzed structural characteristics of curved and rotating disk turbulent boundary layers.
- Coded a module in *Neko* for scalar variance and turbulent scalar flux budgets.

Jan – May 2025 **Group Leader: Multiscale CFD**, *Research Center Pharmaceutical Engineering*, Graz, Austria

2023 – 2025 **Senior Scientist**, *Research Center Pharmaceutical Engineering*, Graz, Austria
- Led the simulation strategy development for an EU Horizon 2020 project.
- Coded a boundary element method from the ground up to resolve sub-Kolmogorov droplet breakup.
  - Unstructured curved elements, Gauss–Legendre quadrature, adaptive mesh refinement, feature-preserving mesh smoothing, adaptive time stepping, matrix-free GMRES(k), fast multipole method.
- Implemented an ethanol-water mixture model in OpenFOAM to simulate impingement jet mixing.

2018 – 2023 **Graduate Research Assistant**, *University of Minnesota*, Minneapolis, MN, USA
*Computational Fluids Lab* (Professor Krishnan Mahesh)
- Performed large-scale ($\sim$10000 processor) simulations of complex turbulent flows using HPC.
  - Formulated streamline coordinates to study curvature in axisymmetric turbulent boundary layers.
  - Resolved experimental trip wires in LES to analyze model-scale boundary layer memory effects.
  - Identified mixing-enhancing secondary vortices from DNS and DMD of a jet in crossflow.
- Extended a massively-parallel in-house unstructured finite-volume overset LES/DNS code.
  - Implemented support for *Hypre* GPU solvers with minimized solver matrix reconstruction.
  - Added viscous flux corrections, LES source terms, and overset grid cutting and hierarchy algorithms.

2016 – 2017 **Undergraduate Research Assistant**, *University of Minnesota*, Minneapolis, MN, USA
*Turbulent Shear Flow Lab* (Professor Ellen Longmire)
- Designed tunable-buoyancy spheres to study particle transport in a turbulent boundary layer.
- Manufactured and set up equipment for particle image velocimetry (PIV) experiments.
- Measured the Stokes number dependence of particle restitution coefficients using high-speed imaging.

2014 – 2019 **Chief Engineer & Aerodynamics Designer**, *University of Minnesota Formula SAE*
- Led weekly meetings of a 70-member team and large-scale ANSYS CFX simulation development.
- Programmed a Matlab graphical user interface to parameterize multi-element wing profiles.

## Service

Co-organizer and session chair of a workshop at the *Banff International Research Station*: "Particulates across Scales: Mathematical Modeling, Computation, and Applications"

Reviewer for the *Journal of Fluid Mechanics* and *International Journal of Heat and Fluid Flow*.

## Teaching

Led wind tunnel labs for laminar/turbulent boundary layer theory and developed and taught recitation exercises for a graduate-level turbulence course.

## Publications

### Journal articles

**N. Morse**. Insights into sub-Kolmogorov-scale droplet breakup via boundary element simulations. *International Journal of Multiphase Flow*, 105460, 2025.

**N. Morse** and K. Mahesh. Tripping effects on model-scale studies of flow over the DARPA SUBOFF. *Journal of Fluid Mechanics*, 975:A3, 2023.

**N. Morse** and K. Mahesh. Effect of tabs on the shear layer dynamics of a jet in cross-flow. *Journal of Fluid Mechanics*, 958:A6, 2023.

**N. Morse** and K. Mahesh. Large-eddy simulation and streamline coordinate analysis of flow over an axisymmetric hull. *Journal of Fluid Mechanics*, 926:A18, 2021.

### Conference papers & abstracts

**N. Morse**. Insights into sub-Kolmogorov droplet breakup via a boundary element method. *BIRS Workshop: Particulates across Scales: Mathematical Modeling, Computation, and Applications*, 2025.

**N. Morse**, J. Remmelgas, and J. Khinast. A simulation framework for nanodroplet breakup in top-down nanoparticle production. *AIChE Annual Meeting*, 2024.

M. Fenelon, Y. Zhang, L. Cattafesta, **N. Morse**, K. Mahesh, L. Li, and Z. Pan. Optimized timing schemes for multi-pulse shake-the-box particle tracking velocimetry. *AIAA SciTech Forum*, 2023.

**N. Morse** and K. Mahesh. The shear layer structure of a tabbed jet in crossflow. *75th Annual Meeting of the APS DFD*, 2022.

M. Fenelon, L. Cattafesta, Y. Zhang, K. Mahesh, and **N. Morse**. Optimizing dt for MP-STB in particle tracking velocimetry. *75th Annual Meeting of the APS DFD*, 2022.

**N. Morse**, T. Kroll, and K. Mahesh. Large-eddy simulation of submerged marine vehicles. *Proceedings of the 34th Symposium on Naval Hydrodynamics, Washington, DC*, 2022.

**N. Morse** and K. Mahesh. Streamline coordinate analysis of the flow past an axisymmetric body computed by large-eddy simulation. *74th Annual Meeting of the APS DFD*, 2021.

**N. Morse** and K. Mahesh. Large-eddy simulation of appended submerged vehicles using an unstructured overset grid method. *73rd Annual Meeting of the APS DFD*, 2020.

T. Kroll, **N. Morse**, W. Horne, and K. Mahesh. Large eddy simulation of marine flows over complex geometries using a massively parallel unstructured overset method. *Proceedings of the 33rd Symposium on Naval Hydrodynamics, Osaka, Japan*, 2020.

**N. Morse**, W. Horne, and K. Mahesh. Towards large-eddy simulation of maneuvering vehicles using an unstructured overset grid method. *72nd Annual Meeting of the APS DFD*, 2019.

D. Barros, Y. H. Tee, **N. Morse**, B. Hiltbrand, and E. Longmire. Investigation of particle lift off in a turbulent boundary layer. *70th Annual Meeting of the APS DFD*, 2017.

## Interests

Mountain biking, road cycling, skiing, running, hiking, tennis, bouldering